

Gradients with a numerical check

Project: Image-pattern Network Investigation

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Slopes and local change

Without opening the old lesson, explain one key idea from Slopes and local change. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Parameters and architecture

Without opening the old lesson, explain one key idea from Parameters and architecture. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Activation functions

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Catch a denominator error for Gradients with a numerical check. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Gradients with a numerical check. Proposed explanation: “The gradient is one scalar”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Check a new example independently

Use $x = 3$, target 2 and $y = wx + b$ at $w = 0$, $b = 1$. With step 0.1, calculate both numerical partials. Compare with [2 times residual times x , 2 times residual].

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Recover a gradient

For the lesson model at $w = 1$ and $b = 1$, calculate prediction, residual, loss and gradient.

My reasoning and evidence

6. W2 Catch a denominator error

A w probe has losses 1.44 and 0.64 at 0.9 and 1.1 with b fixed. A learner divides their difference by 0.1. Correct the result.

My reasoning and evidence

7. W3 Separate a direction from a coordinate

Starting at $[w, b] = [1, 0]$, both coordinates move by plus or minus 0.1 together. Compute the symmetric slope and explain why it is not either partial derivative.

My reasoning and evidence

8. W4 Check a new example independently

Use $x = 3$, target 2 and $y = wx + b$ at $w = 0$, $b = 1$. With step 0.1, calculate both numerical partials. Compare with $[2 \text{ times residual times } x, 2 \text{ times residual}]$.

My reasoning and evidence

9. W5 Recognise a corner

Compute the symmetric difference of ReLU at zero with step 0.1. Explain why this does not prove a derivative exists there.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled gradients with a numerical check evidence sheet to Image-pattern Network Investigation. Show how you can estimate one partial derivative while other parameters stay fixed. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Gradients with a numerical check

Check each coordinate of a gradient while holding the other parameter fixed.

Model $y=2w+b$, target 3. At $[w, b]=[1, 0]$, $y = 2$, residual $y\text{-target}=-1$; loss is residual².

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

Gradients with a numerical check

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Check w direction

Analytic derivative $2 \times \text{residual} \times 2 = -4$.



2. Check b direction

Analytic derivative $2 \times \text{residual} = -2$.



3. Numerical check

With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

CHANGE ONE THING

Swap the two gradient components in the program output.

Explain the changed result using the matching lesson.

Why should the numerical checks perturb one coordinate at a time? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

13. Project portfolio milestone

Project: Image-pattern Network Investigation

Current milestone: Make a transparent first version. Keep a trace of how one input becomes an output; explain the rule or learned component.

For your Image-pattern Network Investigation, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check each coordinate of a gradient while holding the other parameter fixed.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2.

1. Numerical check
2. Check b direction
3. Check w direction

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Numerical check With step 0.1: w losses 0.64, 1.44 give -4; b losses 0.81, 1.21 give -2. A gradient lists the slope for each parameter, in a declared order. Changing w by a small amount changes the prediction twice as much as changing b by the same amount. That explains the different derivative magnitudes. Perturb one coordinate while holding the other fixed, then compare the numerical slope with its analytic counterpart.

Practice 2: explain the deciding evidence

Changed-case practice: Swap the two gradient components in the program output. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The coordinate-wise numerical check exposes the mismatch.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The coordinate-wise numerical check exposes the mismatch. The w perturbation changes y twice as strongly as the same b perturbation.

Repair a missing building block

Review Slopes and local change (ai-5/9). Goal: Calculate an average rate of change over a stated interval. Short practice: Which calculation disproves “any increase helps”? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Parameters and architecture (ai-5/7). Goal: Distinguish architectural choices from parameter values. Short practice: How many new parameters did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Review Activation functions (ai-5/4). Goal: Trace identity ReLU and sigmoid on a numerical score. Short practice: Which original inputs cannot be distinguished by this network’s final output? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
